1

**MA3521 — $\mathbf{y}' = A\mathbf{y} + \mathbf{f}(t)$ Exponential and Trig (I)**

Name:

1. Solve $\mathbf{y}' = A\mathbf{y} + e^{-2t}\begin{pmatrix}-1\\-5\end{pmatrix}$, $\mathbf{y}(0) = \begin{pmatrix}2\\3\end{pmatrix}$ and $A = \begin{pmatrix}1 & -2\\2 & 1\end{pmatrix}$ with epairs $1+2i$, $\begin{pmatrix}1\\-i\end{pmatrix}$; $1-2i$, $\begin{pmatrix}1\\i\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$

(b) $\mathbf{y}_p(t) =$ is a suitable UC form.

(c) Row reducing gives the UCs in

$\mathbf{y}_p(t) =$

(d) Row reducing gives constants $\mathbf{c}$ in

$\mathbf{y}(t) =$

2. Solve $\mathbf{y}' = A\mathbf{y} + \cos(2t)\begin{pmatrix}1\\4\end{pmatrix}$, $\mathbf{y}(0) = \begin{pmatrix}1\\-3\end{pmatrix}$ and $A = \begin{pmatrix}1 & -2\\2 & 1\end{pmatrix}$ with epairs $1+2i$, $\begin{pmatrix}1\\-i\end{pmatrix}$; $1-2i$, $\begin{pmatrix}1\\i\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$

(b) $\mathbf{y}_p(t) =$ is a suitable UC form.

(c) Row reducing gives the UCs in

$\mathbf{y}_p(t) =$

(d) Row reducing gives constants $\mathbf{c}$ in

$\mathbf{y}(t) =$